# The Squid Programs: Trading the VIX Curve

Dynamic volatility using the VIX term structure

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A long-standing question in the financial literature has been the search for a well-defined entry point when trading volatility-based strategies and products, as a way to counterbalance negative equity returns during periods of market uncertainty. One of the most important measures of said uncertainty is the VIX Index, which measures the 30-day expected volatility of the U.S. stock market, derived from prices of options on the S&P 500 Index.

In this white paper, we introduce the Squid Programs, a suite of strategies that explore a possible answer to this question by looking at the VIX term structure. The resulting framework dynamically allocates to equity futures and volatility futures, taking inspiration from the S&P 500 VIX Futures Term-Structure Index [1] when allocating to volatility.

## Introduction and Trading Parameters

Volatility strategies are notoriously dependent on path and timing: they can provide powerful convexity during stress, but they often suffer negative carry in calm regimes. The VIX futures term structure provides a compact, observable proxy for determining the current regime. Indeed, in normal markets the curve is typically upward sloping (contango), reflecting the so-called variance risk premium; during stress, the curve tends to flatten, invert, or become locally "scrambled" as near-term protection is bid aggressively. We exploit this intuition by measuring how far today's curve departs from the usual contango ordering, and we use that "dislocation" signal to switch systematically between equity futures and volatility futures.

This white paper is structured as follows: after defining key parameters, we introduce the S&P 500 VIX Futures Term-Structure Index and explain the key concepts regarding its composition. We then compare the return profiles of equity and volatility futures across term-structure regimes. The VIX term structure is taken as discriminant: we summarize its state by counting how many of the first seven VIX futures violate the usual contango ordering UX1 $<$ UX2 $<$ ... $<$ UX7. This dislocation count serves as the key indicator for dynamically allocating between equity and volatility futures.

In this work we will use the following two assets:

Table 1: Assets

| Asset Class | Contract |
|---|---|
| Equity Futures | ES (E-Mini S&P500 futures) |
| Volatility Futures | UX (VIX futures) |

The notation UX1, UX2 etc. refers to the 1st, 2nd etc nearest-expiration contract. The rolling schedule follows the standard procedure for ES: quarterly rolling happens five days before the first notice date, in order to match the active contract (the existing position is rolled in a single trading day). For VIX futures, rolling takes place every month, three days before the date of first notice. Trading is assumed to take place at the close of the regular trading session. Transaction costs in futures are typically modelled as bid-ask spreads plus fixed per-contract fees (commission, exchange, clearing) rather than pure percentage of notional. We use a constant 2 bps all-in assumption as approximation for both UX and ES, and for all trades; unless stated otherwise, all performance statistics and plots reported in this paper are net of these assumed all-in transaction cost. We used data from 2006/04/05 until 2025/12/31.

## The VIX: Index, Futures, Options

Before proceeding, we will look at the VIX in more detail to clarify general notions about this Index. The VIX represents the market's risk-neutral expected

variance of the S&P 500 over the next 30 calendar days, expressed as annualized volatility. In other words, it is the square root of the fair 30-day variance swap rate on the S&P 500, quoted as a percentage standard deviation per annum. For instance, if VIX is 20, this roughly implies a 30-day variance expectation of $20^2 = 400$ in annualized variance units. The VIX is forward-looking, constructed by interpolating between two nearby option maturities so that it consistently reflects a 30-day horizon.

Because the VIX is based on option-implied, risk-neutral expectations, it covers both the market's uncertainty about future volatility and a so-called variance risk premium. This risk premium reflects how option markets price variance, rather than a pure statistical forecast. As a result, VIX levels typically overstate subsequent realized volatility. Realized volatility, on the other hand, looks backward to measure how much the market actually moved over a given period.

Historically, on average, implied volatilities have exceeded realized volatilities: the phenomenon where implied variance, and thus VIX, exceeds realized variance on average is widely documented [Carr et al., 2009], [Bollerslev et al., 2009], [Bekaert et al., 2014], or [Dew-Becker et al., 2023]. It arises from investors' willingness to pay for insurance against large market moves: sellers of variance or volatility demand compensation for bearing this tail risk, resulting in market prices that systematically cover this premium over expected outcomes. This does not necessarily imply that there is a mispricing; rather, it is an equilibrium compensation for risk transfer.

We previously defined the VIX in terms of variance swaps. A variance swap is an over-the-counter forward contract on future realized variance. The payoff to the long side at expiration is proportional to the difference between the actual realized variance and a predetermined strike variance. Let us define $K_{\text{var}}$ as the agreed strike (usually quoted in volatility terms) and $N$ as the notional. The buyer receives a payoff $P$ equal to

$$P = N \times (V_{\text{realized}} - K_{\text{var}}), \tag{1}$$

where $V_{\text{realized}}$ is the realized variance. Because the underlying measure is variance, the payoff profile is nonlinear in terms of volatility. A doubling of volatility (e.g. from 20% to 40%) quadruples variance (0.04 to 0.16), which gives amplified gains or losses, adding a whole new dimension to tail risk for sellers. Suppose a fund sells a 2-month variance swap struck at 16% implied volatility (variance $= 0.16^2 = 0.0256$). If realized volatility over that period actually turns out to be 32% (variance = 0.1024), the payout the seller owes is

$$N \times (0.1024 - 0.0256) = 0.0768 \times N.$$

This convex payoff amplifies outlier market moves, potentially causing large losses for the seller. This is the main reason why an equity index variance swap is usually priced above the corresponding implied volatility at-the-money. It is therefore misleading to compare the VIX with realized volatility and draw conclusions about the risk premium without accounting for this extreme tail risk.

Because the VIX itself is not a tradable index[1], a way to get exposure to the VIX is via VIX futures. VIX futures provide a linear payoff on future implied volatility levels. Buying a 6-month VIX future at 18, for instance, yields a point-for-point gain or loss relative to the final settlement value of the VIX at expiration. If the VIX settles at 36, the profit is $36-18=18$ (times the contract multiplier). VIX options are priced off the corresponding VIX futures, not the spot VIX Index. VIX options cash-settle to the special opening quotation (SOQ) of the VIX on expiration, which itself is derived from SPX option prices that now have 30 days to maturity. This means that each VIX option maturity corresponds to a particular VIX futures contract.

VIX futures are usually (meaning during normal market environments) in contango, a situation where longer-dated expirations trade at higher levels than near-dated contracts, in order to reflect the volatility risk premium. VIX options inherit the same structure as a consequence of being priced off futures. A 4-month call option on VIX is priced with respect to the UX4, which should trade a few points above the spot VIX (in a normally-functioning market). Thus, if the spot VIX rises from 14 to 30 but the 4-month future only rises from 17 to 24, a call option on that 4-month future sees a smaller gain than one might expect by looking at the spot index alone.

Another point worth emphasizing concerns the origin of the variance risk premium and its relation to the shape of the SPX volatility surface. As noted above, implied volatility has historically exceeded realized volatility on average, reflecting investors' willingness to pay for protection against large adverse market moves. In practice, this demand is concentrated in downside insurance, giving rise to the persistent left-tail skew of index options. Because the VIX is constructed from a strip of SPX options across strikes, it naturally incorporates this skewness. The variance risk premium should therefore be understood not only as a gap between implied and realized volatility, but also as the equilibrium price of structurally demanded crash protection.

Relatedly, the term structure of volatility may be viewed as the outcome of recurring and maturity-specific order flow in the index options market. Shorter maturities are often influenced by systematic option-selling activity, including call overwriting, while intermediate maturities tend to reflect more persistent hedging demand for puts. Farther out the curve, the hedging of structured products may create net vega supply, which can act to flatten longer-dated maturities. The overall shape of the curve therefore is also a reflection of equilibrium supply and demand across horizons.

[1]It is possible in principle to replicate the VIX via options and trade it, and this has been explored in the literature, see e.g. [Demeterfi et al., 1999]. Realized variance can be replicated by a static portfolio of European calls and puts. If one were to purchase all the options used in the VIX calculation, however, the position would then be subject to PnL fluctuations due to option decay and actual movements in the underlying asset. By the next day, the basket of options would no longer match the original composition, requiring rebalancing. The VIX simply reflects the cost of the current basket of options; it doesn't indicate whether owning that basket will be profitable.

## The S&P 500 VIX Futures Term-Structure Index

We briefly recall the main characteristics of the S&P 500 VIX Futures Term-Structure Total Return Index (we will use the Bloomberg ticker SPVXTSTR) and refer the reader to the original documentation [1] for details.

The SPVXTSTR is a hybrid volatility strategy, combining a long position in mid-term VIX futures with a short position in short-term VIX futures. Specifically, it takes a 100% long position in the S&P 500 VIX Mid-Term Futures Index and a 50% short position in the S&P 500 VIX Short-Term Futures Index:

$$\text{SPVXTSTR} = -0.5\ \text{Short-Term Index} + 1\ \text{Mid-Term Index} \tag{2}$$

These two indices are further detailed in [1], but they basically amount to maintain continuous rolling positions in VIX futures to target constant maturities. The Short-Term holds the 1st-month and 2nd-month VIX futures, rolling a fraction of the position each day from the nearest contract (UX1) into the next (UX2) to keep an effective one-month maturity. Similarly, the Mid-Term holds the 4th, 5th, 6th, and 7th month VIX futures. It starts each rolling cycle with equal weights in those four contracts, then each day it sells a fraction of UX4 to buy UX7, without touching UX5 and UX6. Over the course of a month, the index gradually shifts its allocation out of the expiring UX4 into UX7, which will become UX6 at the next rolling date. This daily rolling process gives rise to roll yield, which is a key component of the index's performance.

The index is structured in such a way that positive returns arise in two ways, related to contango and backwardation. In a contango market, rolling a long position to a farther month costs money (by definition). This drag on returns results in a negative roll yield. The SPVXTSTR therefore produces returns by shorting the roll on the front end of the curve, which is usually the steepest part, earning positive carry. By the same principle, the other source of returns comes from the long end of the curve during times of backwardation. Obviously, the positive carry earned by the front end will be losing when that part of the curve is backwardated, just as the long end position will lose in contango markets. The balance between these two opposite sources of positive and negative returns keeps the index positive on average.

## Returns distributions

As seen above, the VIX futures curve normally operates in contango. During market stresses, however, concerns about immediate volatility can outweigh the volatility risk premium that steepens the curve, making it flat or even inverted. A measure of stress can therefore be identified in the deviation from the "natural" state of the market, i.e. UXn $<$ UX(n+1) $\forall$ n. We will consider from now on maturities up until n = 7, as in the SPVXTSTR. A state of perfect contango is therefore:

$$\text{UX1} < \text{UX2} < \text{UX3} < \text{UX4} < \text{UX5} < \text{UX6} < \text{UX7}. \tag{3}$$

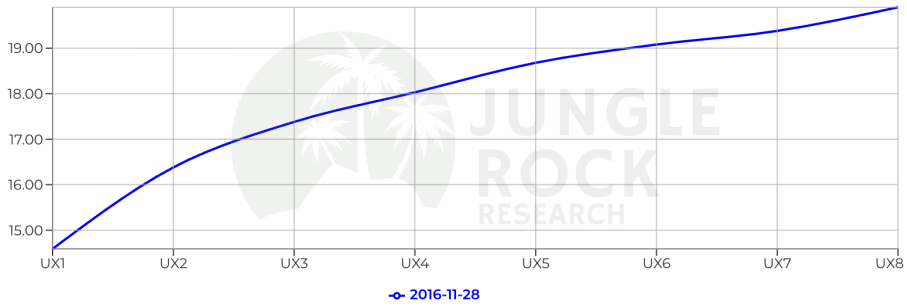



Figure 1: VIX term structure in contango on the date 2016-10-12.

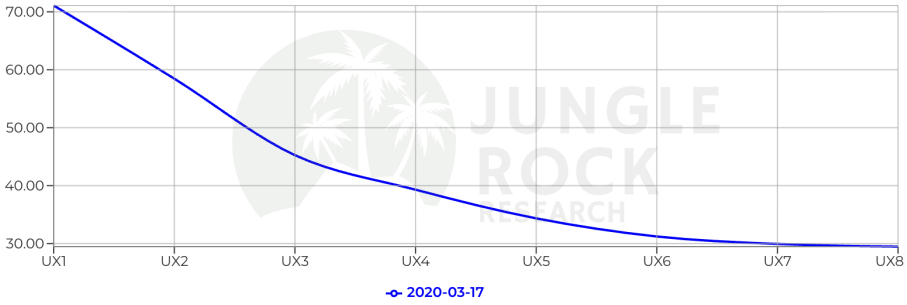



Figure 2: VIX term structure in backwardation during a period of market stress: we took the date 2020-03-17, during the Covid pandemic.

A rough measure of the degree to which the market is not operating smoothly is the number of deviations from Eq. (3). A relative short-term stress could be typically signalled by an inversion of the first two maturities, UX2 $<$ UX1. A more structural breakdown could involve a inversion of, say, four contracts: UX4 $<$ UX3 $<$ UX2 $<$ UX1. Extreme flattening of the curve or pricing of later events where volatility is expected to soar might give rise to more complex configurations where the curve is neither fully inverted nor in contango, so this measure is not always correct. In general, however, the number of deviations from Eq. (3) can be considered a good proxy for the state of the market. We will get into more details in the following sections.

We can quantify this by looking at the distribution of S&P 500 Index returns. We sample the average return in three buckets: i) days of perfect contango, ii) days of low dislocation (up to four dislocations) and iii) days of high dislocations (more than four dislocations). Figure 3 shows a bar chart representing these quantities.

We can see that returns for the S&P 500 are maximized during days of perfect contango. Days with a low number of dislocations show a decrease, but returns are still positive. For high dislocations, the returns are large and

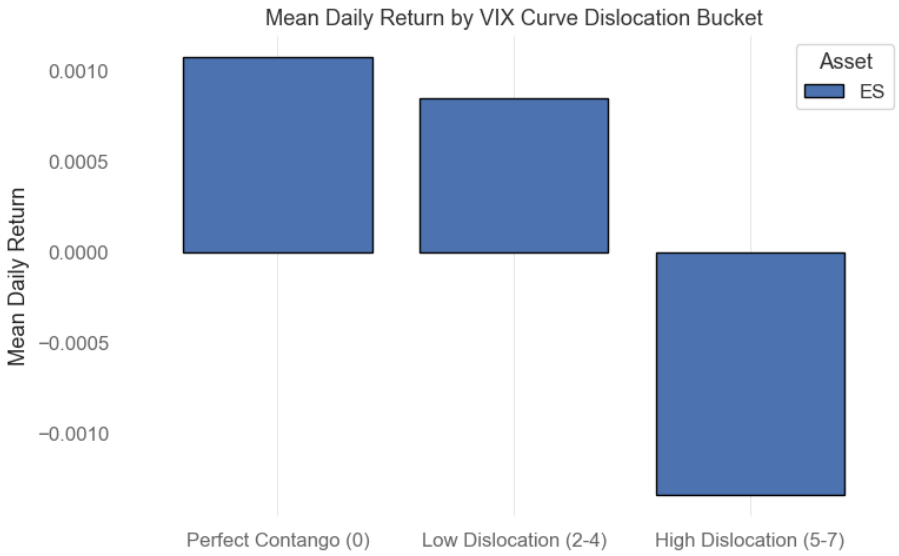



Figure 3: Average returns of ES for the three different regimes of VIX term structure dislocations.

negative.

When looking at the same plot for the SPVXTSTR the situation is very different, as seen in Figure 4, where we show a bar plot comparison between ES and the SPVXTSTR. The two components of the index manages to balance each other well in all dislocations regimes, with the maximum average return achieved in days with a low number of dislocations, followed by (smaller) positive return in days of perfect contango. Most crucially, however, we see a small positive average return achieved in days with a large number of dislocations as well, in contrast to what happens for ES.

Table 2 gives the summary statistics. The days of perfect contango are the most numerous, as expected given our initial discussion: the market in a 'normal" state prices a volatility/variance risk premium, which is reflected in the term structure as contango .

| Bucket | Count | ES avg return | SPVXTSTR avg return |
|---|---|---|---|
| Perfect Contango (0) | 2761 | 0.108 % | 0.040 % |
| Low Dislocation (2-4) | 887 | 0.085 % | 0.131 % |
| High Dislocation (5-7) | 1311 | -0.134 % | 0.003 % |

Table 2: Summary statistic for the return distributions, sampled according to the number of dislocations in the VIX term structure.

A strategy that dynamically allocates between the two indices based on the number of dislocations would therefore capture most of the upside during perfect

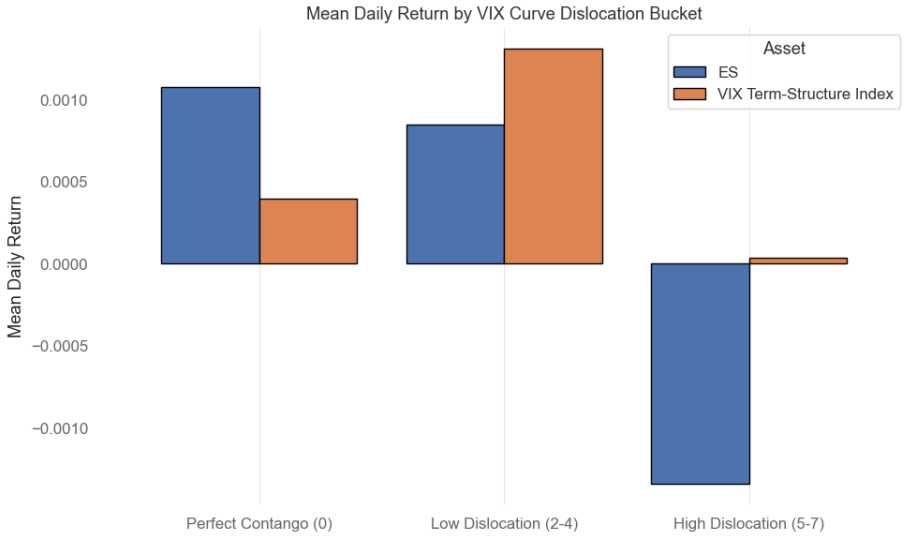



Figure 4: Average returns of SPVXTSTR and ES for the three different regimes of VIX term structure dislocations.

contango markets, when it is fully allocated to the S&P 500, without suffering very large losses when the VIX curve is deviating from Eq. (3), by allocating to the SPVXTSTR.

This is the essence of the Squid Programs: using the number of dislocations as a way to weigh between two components, namely equity and volatility.

## Strategy composition

In order to build a strategy, we formalize the idea of a rank indicator as the normalized sum of the number of deviations from Eq. (3).

$$w = \frac{1}{7}\sum_{j=1}^{7}\begin{cases}1 & \text{if} \quad |\{\, i : \mathrm{UX}i < \mathrm{UX}j\}| + 1 \neq j \\ 0 & \text{otherwise}\end{cases}, \tag{4}$$

where the notation $|\{\,\}|$ indicates the cardinality of the set. In other words, we are checking that the sorted sequence of $\{\mathrm{UX}j\}$ corresponds to the numerical ranking, and counting the number of items that are out of place.

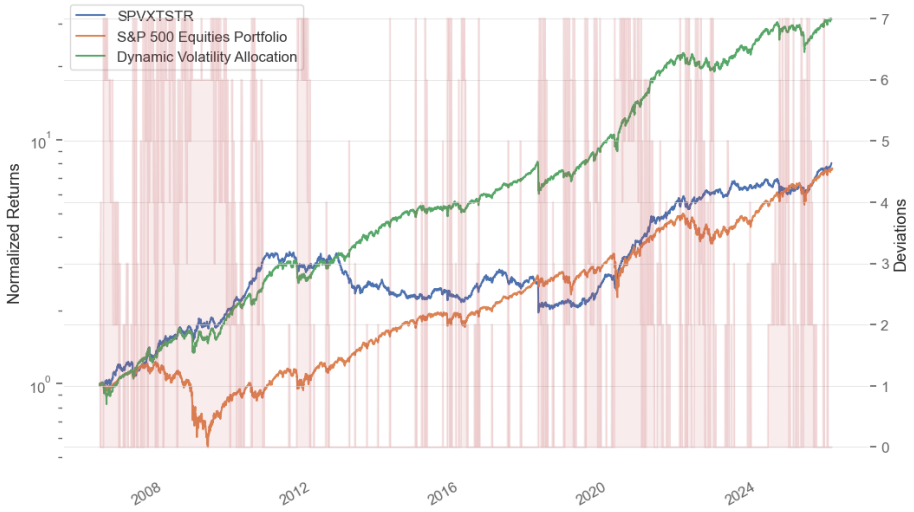
The weight $w$ is the percentage allocation to ES. It is maximized when the condition in (4) holds for all $j$s, giving $w = 1$; conversely, we have $w = 0$ when the condition doesn't hold for any $j$. We compose an initial strategy $\mathrm{S}_0$ as:

$$\mathrm{S}_0(t) = w_{(t-1)}\ \mathrm{ES} + (1 - w_{(t-1)})\ \mathrm{SPVXTSTR}, \tag{5}$$

where $w_t$ represents the weight $w$ of Eq. (4) for a given day $t$.

In Figure 5 we plot the results for $S_0$, labeled as Dynamic Volatility Allocation in green. On the right Y-axis we show the number of deviations in red-shaded area. As expected, in periods of market turmoil, such as the Great Financial Crisis or the Covid Pandemic, the number of deviations is very high, while it is very low in e.g. the years between 2013 and 2015. The green line outperforms both components by dynamically shifting between them, allocating to each one during periods when it is more advantageous to do so, based on a simple indicator.

Figure 5: Plot of $S_0(t)$ and of its two components.




We could already be satisfied with this result. Instead of allocating directly to the SPVXTSTR, however, we will introduce a few adjustments, making considerations about the key performance drivers of this Index.

From our previous discussion we can see that returns are mainly generated by a short position in the front end, coupled with a long position in the long end of the curve. However, futures in the long end are much less liquid with respect to the front end, and this poses issues in terms of scalability. We apply therefore this crude approximation: replace the short component in Eq. (2) with just UX1, and replace the long component with just UX3, keeping the allocation weights $\{-0.5, 1\}$. Therefore, we have a volatility allocation that closely resembles Eq. (2) but only uses two maturities:

$$\text{VA} = -0.5\ \text{UX1} + \text{UX3}, \tag{6}$$

where VA stands for Volatility Allocation. We arrive then to a new strategy $S_1$:

$$S_1(t) = w_{(t-1)}\ \text{ES} + (1 - w_{(t-1)})\ \text{VA}. \tag{7}$$

In the above equation (7), we are not trading a constant-maturity exposure anymore: indeed, we are not rolling a combination of UX1 and UX2 in order to have a one-month synthetic VIX future exposure, as in the SPVXTSTR. We are now holding UX1 instead, and rolling it according to the chosen rolling schedule (same argument goes for the Mid-Term component and UX3). This means that we inherit curve risk and that the reactivity of the futures may vary depending on the vicinity of the expiration date. At the same time, we are sure that liquidity does not create an issue when trading the strategy and we greatly simplify strategy management as an appealing side-effect.

Figure (6) shows the difference between the two volatility allocations alone. While the overall performance of VA is clearly worse, during periods with high numbers of deviations the performance picks up, making sure that this substitution does not greatly impact the full strategy in Eq. (7).

Not having a fixed-maturity requirement allows us to consider trading frequencies other than daily. After all, even if our initial analysis with division in buckets was done on daily returns, we can argue from inspection of Figure 5 that period of market stresses can persistently show curve dislocations. We move therefore from a daily to a weekly model, and take the weekly average number of deviations in our calculations of $w$. Moving from daily to weekly rebalancing materially reduces turnover (see Table 3), further improving implementability. The resulting weekly program is the Squid Program, shown in Figure 7. We call Cuttlefish the daily version.

Table 3 shows the summary statistics of the Squid and Cuttlefish Programs compared to the S&P 500.

Table 3: Perfomance statistics.

| Variable | S&P 500 Index | Weekly Program: Squid | Daily Program: Cuttlefish |
|---|---|---|---|
| Annualized Returns | 10.77% | 15.84% | 17.59% |
| Annualized Volatility | 17.26% | 13.49% | 13.42% |
| Max Drawdown | 55.19% | 24.09% | 30.10% |
| Sharpe Ratio | 0.54 | 1.06 | 1.20 |
| Sortino Ratio | 0.44 | 1.18 | 1.42 |
| Turnover | N/A | 4.46% | 15.45% |




Figure 6: Comparison of the volatility allocation parts in $S_0(t)$ and $S_1(t)$.

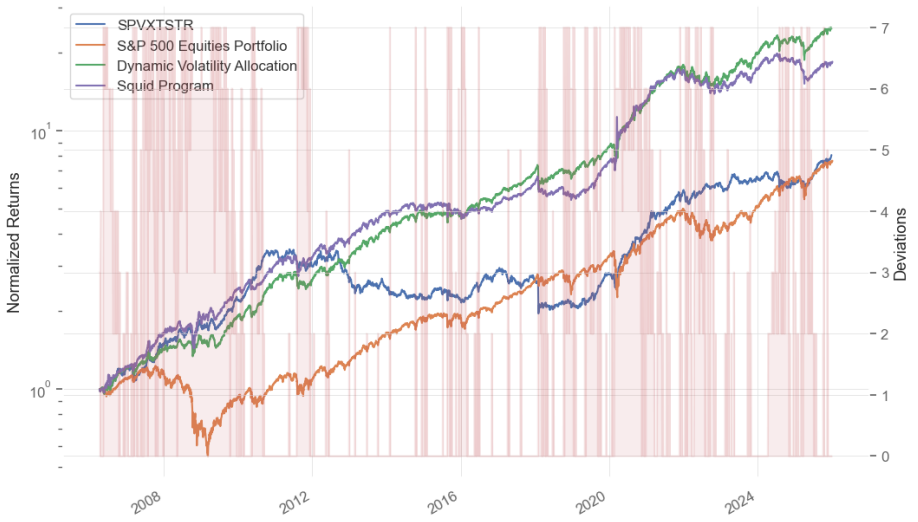



Figure 7: The Squid Program compared $S_0(t)$ and its components.

## Refining the strategy

The number of deviations from contango is our key driver in deciding the allocations of Squid and Cuttlefish.

This measure is, nevertheless, imperfect. The attentive reader might have noticed this when we first introduced the concept, but let us expand on this matter in order to refine the strategy. A fully backwardated curve, by construction, will have a lower number of dislocations (namely six, as UX4 keeps the same position in the inequality and does not contribute in Eq. (4)) with respect to a flat curve where maturities happen to be scrambled in a few points around a single value. In theory, the first case represents the most stressed possibility, but we are giving it a lower weight in terms of our measure.

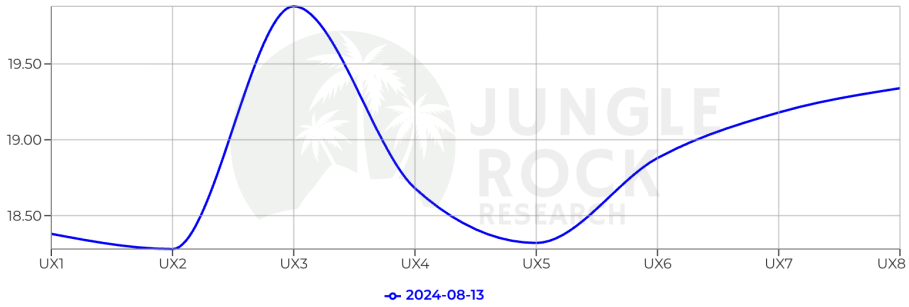



Figure 8: The VIX term structure may appear scrambled, not fully backwardated nor in contango, within a very small range of values. We show here the date 2024-08-13; note the values on the vertical axis, with the whole curve ranging between two points.

We can therefore make a first improvement by introducing the concept of slope. Fully backwardated curves, with high negative slopes, should have larger volatility allocations compared to mostly flat and scrambled curves. We may define the slope $m$ as follows:

$$m = 1 - \frac{\text{UX1}}{\text{UX7}}. \tag{8}$$

For values above a critical (negative) $m_c$ value, but still less than 0, we choose to dampen the volatility contribution by 1/2. This will make sure that only large, negative slopes will get the full VA contribution, and inverted but flatter, scrambled curves will get only half the contribution.

$$\text{S}_1(t) = \begin{cases} (\frac{1}{2}w_{(t-1)} + \frac{1}{2})\ \text{ES} + \frac{1}{2}(1 - w_{(t-1)})\ \text{VA}, & 0 > m > m_c, \\ w_{(t-1)}\ \text{ES} + (1 - w_{(t-1)})\ \text{VA} & \text{otherwise.} \end{cases} \tag{9}$$

However, this is still a relative measure. From our initial discussion about the variance, we know that the VIX Index carries information as an absolute

measure of expected variance as well. High-vol environments (large VIX values) can therefore carry implicit risk even if the curve exhibits a small, overall positive slope $m > 0$ with dislocations in between (see e.g. Figure 8). To take this risk into account, we may implement a filter on a threshold VIX level. When above this threshold $V_c$, we double the weight of the volatility component VA of Eq. (7). As the total weight must trivially sum to 1, this means that (when $m > 0$) one has:

$$\mathrm{S}_1(t) = \begin{cases} (2w_{(t-1)} - 1)\ \mathrm{ES} + 2(1 - w_{(t-1)})\ \mathrm{VA}, & \mathrm{VIX} > V_c, \\ w_{(t-1)}\ \mathrm{ES} + (1 - w_{(t-1)})\ \mathrm{VA} & \text{otherwise.} \end{cases} \tag{10}$$

The role of these thresholds, $m_c$ and $V_c$, is to define broad regime transitions; the strategy logic does not depend on fine-tuning them to a narrow optimum. Such a configuration naturally allows to explore the possibility of taking tactical shorts[2]. This is because in Eq. (10) we impose coefficient weights trivially summing to 1 in front of the components. It follows that, should the VA weight become larger than 1 after doubling it, we must have $2w_{(t-1)} - 1 < 0$, hence a short position for ES.

Bringing everything together yields a new weekly program, which we call Giant Squid, along with its daily counterpart, Colossal Squid. If we remove the ability to short ES from Giant Squid while retaining all other enhancements, we obtain a third program: Jumbo Squid. The corresponding equity curves are shown in Figures 9 through 12, and the performance statistics are reported in Table 4.

Table 4: Perfomance statistics.

| Variable | S&P 500 Index | L/S ES Weekly Giant Squid | Long ES Weekly Jumbo Squid | L/S ES Daily Colossal Squid |
|---|---|---|---|---|
| Ann. Returns | 10.77% | 21.01% | 19.40% | 21.09% |
| Ann. Volatility | 17.26% | 15.34% | 13.72% | 14.90% |
| Max DD | 55.19% | 24.14% | 24.09% | 29.71% |
| Sharpe Ratio | 0.54 | 1.27 | 1.30 | 1.31 |
| Sortino Ratio | 0.44 | 1.69 | 1.60 | 1.73 |
| Turnover | N/A | 8.23% | 6.08% | 26.59% |

[2]In appendix A we show that ES has large returns at $t + 1$ in the high dislocation bucket, so it makes sense to take shorts only for small positive slopes.

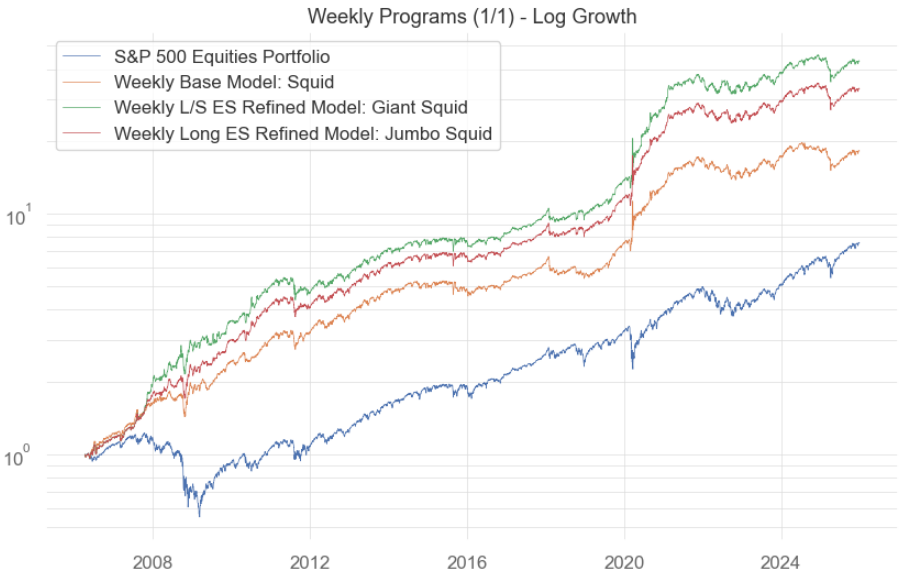



Figure 9: Weekly Programs. Log growth.

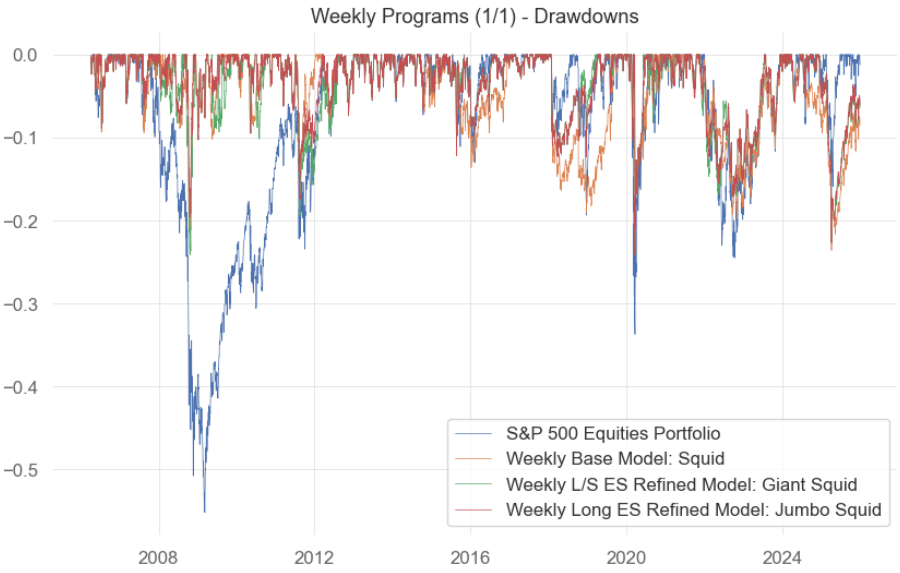



Figure 10: Weekly Programs. Underwater plot.

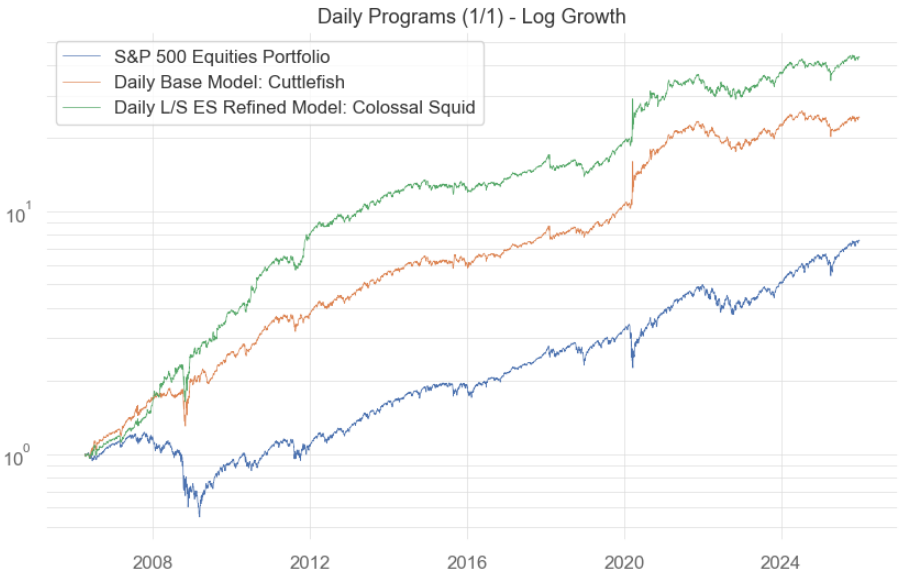



Figure 11: Daily Programs. Log growth.

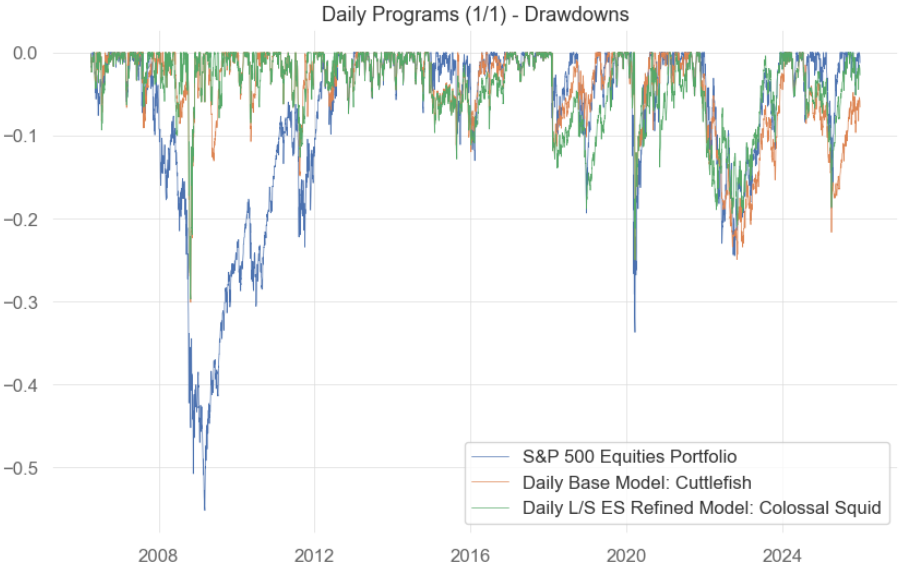



Figure 12: Daily Programs. Underwater plot.

## Discussion and outlook

There are a few points worth expanding on, regarding the methodology employed above. We based our strategy on the number of deviations from contango in the VIX futures term structure. In order to make that choice, we looked at dislocations grouped in three buckets and checked the returns on the same day. However, when implementing the strategy, we are using the term structure at $t-1$ in order to allocate at $t$. We discuss this issue in detail in Appendix A.

Further improvements to the model should take into account the curve as a whole, while also examining each section individually to identify where dislocations occur. A measure of the amplitude of such dislocations could provide a more accurate signal: a single, very small deviation, where say UX6 < UX5, is currently treated in much the same way as a much larger one, where one might have UX2 < UX1, even though the impact and interpretation of the two are evidently quite different. Targeting the specific parts of the curve that are dislocated may also create opportunities for deciding where to allocate, rather than always using only UX1 and UX3 as done above. Previously, slope was estimated using only UX1 and UX7. One possible refinement would be to compute the actual slope using all seven points; since this can be done every day, its distribution could then be used to gauge the degree of deviation, or discordance, from the normal contango structure. The system suffers most when the curve remains flat for long periods while also presenting a large number of dislocations, and such an approach could help avoid negative contributions from these occurrences.

Expanding on the rationale behind the strategy, it is worth to go back to a point that was raised above regarding options flows. The structural order flow in index volatility products is non-trivial, and it likely reinforces the regime condition that the strategy seeks to capture. In particular, a term structure in contango implies a natural rolldown, that is, a tendency for fixed-maturity implied volatility exposures to drift lower as expiration approaches. Such a drift may also induce a modest but persistent rebalancing flow in the underlying hedge instruments. As long-call and short-put exposures decay, their absolute deltas decline, and dealers hedging these positions may need to buy back part of the hedge asset in order to remain delta-neutral. In our setting, where ES futures are the relevant hedge instrument, this mechanism may contribute to a subtle supportive flow in calmer regimes. While one should be careful not to overstate the magnitude of this effect, it offers a plausible microstructural view through which contango, option flows, and the underlying index may interact.

Another important aspect for our purposes is the distribution of positioning across strike, maturity, and product type, and the way in which that positioning shapes the skew and the term structure. At the short end, heavy option selling, often dominated by overwrite activity, can suppress a large portion of the volatility surface outside the tails, with call implied volatilities particularly depressed. Farther out, hedging demand can become relatively more important and may even imply net vega buying by end users, reinforcing an upward-sloping curve. In this sense, the observed distribution of option supply and demand, with calls

more readily supplied and puts more persistently demanded, is consistent with the drift that the strategy is attempting to exploit. This does not by itself prove the validity of the signal, but it does provide a structural justification for why the term structure may contain useful information.

Another point to raise is about the dataset that we are looking at. Efficiency has greatly improved in the years and listed VIX futures are quite recent (reliable data starts from 2006). A key driver of the total performance is indeed the period during the Great Financial Crisis: volatility stayed elevated, with massive spikes, generating returns from the VA part of the strategy. In contrast, the strategy struggled the most and suffered its greatest drawdowns during the 2022 bear market and the 2025 tariffs crash. In both cases, volatility was not as wild but sticky, the VIX curve remained flat for long periods, with the VIX rarely breaking over 35 or below 20. This may also be attributed to a market moving more efficiently in shorting volatility compared to 2008. In such a situation, it is possible that implementing the changes discussed above, to give a better measure of the allocation weight signal, would help greatly.

Another very interesting question is to extend the model to intraday signals, but it is outside the scope of this white paper.

# Conclusion

We have shown that the VIX term structure can be used as a tool to dynamically allocate between equity and volatility. In normal regimes the VIX futures curve is in contango, and departures from this monotone shape, or dislocations, are a practical proxy for stress. The Squid Program uses the resulting dislocation signal to dynamically switch between equity futures and volatility futures. Specifically, our measure was based on counting the number of deviations from a state of contango in the VIX term structure. This measure acted as weight in allocating between equity futures and volatility futures. We initially took the SPVXTSTR as candidate for our volatility allocation, modifying afterwards the strategy based on liquidity and turnover considerations. We arrived eventually to a weekly strategy that dynamically switches between equity and volatility: the Squid Program. We have shown that this program achieves attractive returns, Sharpe ratio and drawdown profile by means of its dynamical adjustments based on the core signal. From this starting point, we have constructed other programs by taking into account other measures, such as the magnitude of the deviations themselves, the VIX level, the possibility of shorting ES, and switching back to daily signals. Further work in this direction, outlined in the Discussion section, is expected to deliver even more interesting results.

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[1] `https://www.spglobal.com/spdji/en/documents/methodologies/methodology-sp-vix-futures-indices.pdf`

[2] `https://www.volsignals.com`

# Appendix A

Figure 4 shows the average returns for three different buckets of the current state of the VIX term structure. This means that we looked at the measure of the curve and the returns on the same day. In practice, however, it is clear that no such thing is achievable if one wants to use a daily signal for trading, and we must use the shifted value in order to allocate, as we did in Eqs. (5) and (7). We expand now the above reasoning and look at the shifted distributions. In Figure 13 we show the grouping in buckets using the returns at $t + 1$.

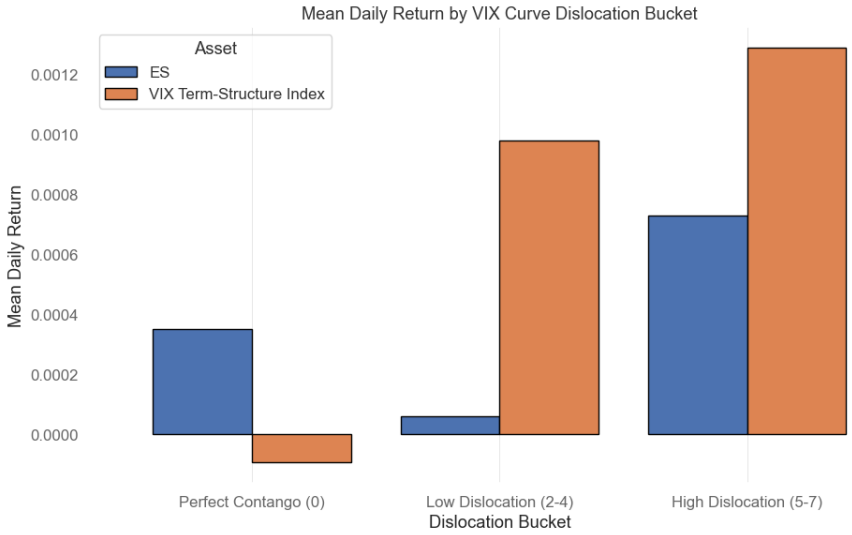
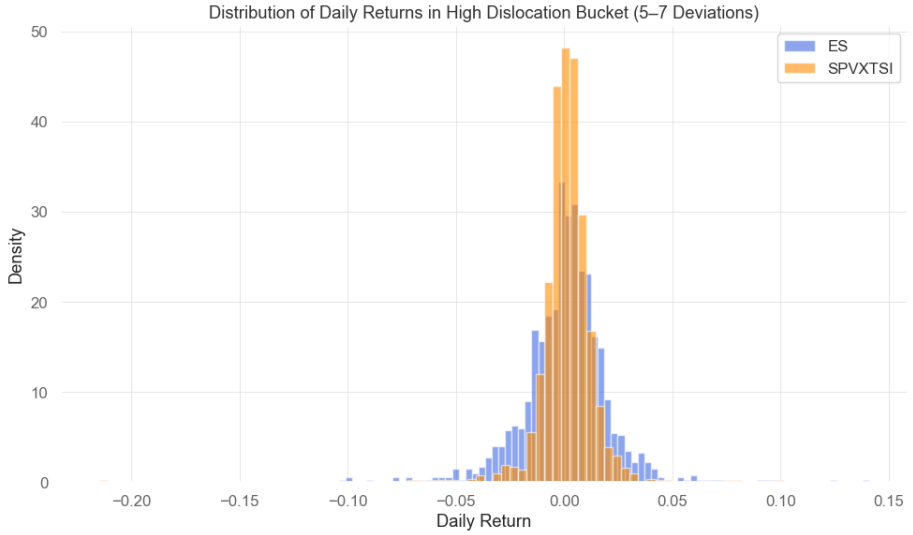



Figure 13: Average returns at $t+1$ of SPVXTSTR and ES for the three different regimes of VIX term structure dislocations.

The results confirm our findings: ES outperforms during contango, while the SPVXTSTR outperforms elsewhere. The small positive returns of the SPVXTSTR in contango become negative when looking at $t + 1$, even more reason to avoid allocating to the volatility component in contango. Perhaps the surprising result is the behavior of ES, with the other two buckets passing from large positive to almost zero and from negative to positive. Further investigation in the distributions of the last bucket gives the following two plots.

Figure 14: Distributions of the daily returns in the high dislocation bucket at $t+1$.



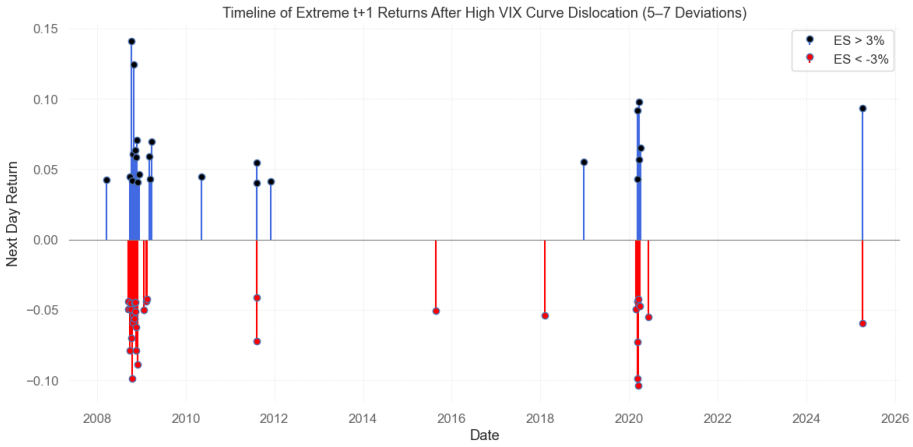




Figure 15: Stem plot of outsized returns in the high dislocation bucket at $t+1$ for ES.

Figure 14 shows the distribution of returns for the last bucket, while Figure 15 shows oversize returns in the distribution (above 3% changes in either direction), just for ES, in a stem plot with the dates on the horizontal axis.

We clearly see that the distribution of the SPVXTSTR is much more peaked with thinner tails. It is instructive to look not only at the mean, but at the median as well, as shown in Table 5. For the SPVXTSTR, the two are much closer, pointing towards a robust result. ES shows instead a much higher median with respect to the mean, indicating therefore a negative skew, and dependence on tail behavior. Looking at the stem plot, we can easily identify the components. Outsized positive returns are mostly part of bear market rallies and periods of recovery, from which we get positive contributions. Overall, though, the fact that the mean gets pulled to the left of the distribution with respect to the median indicates potential negative drag on returns, and this contingency should be avoided in favor of a more stable result such as the one seen for the SPVXTSTR. This further supports our initial analysis: as it is also confirmed by the resilience of the strategy when moved over to a weekly signal, the signal at $t-1$ still serves it purpose as a measure of instability and it is a well-behaved indicator to trade at time $t$.

| Asset | Median | Mean |
|---|---|---|
| ES | 0.1569 % | 0.0729 % |
| SPVXTSTR | 0.1362 % | 0.1288 % |

Table 5: Statistics for the distributions of the $t+1$ returns in the high dislocations bucket.